EE4047 – High Voltage Techniques I

Continuation to:
Generation of High Voltages

Week 4.2 – Thursday Lesson

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OUTLINE

3 Continuation to: Alternating Voltages

3.1 Continuation to: Testing transformers

3.2 Series resonant circuits

4 Impulse Voltages

3 ALTERNATING VOLTAGES

The winding ‘3’ supplies the primary of the second transformer unit II, both windings are dimensioned for the same low voltage and the potential is fixed to the high potential V. The secondary windings ‘2’ of both units are series connected so that a voltage of 2 V is produced hereby. The addition of the stage III is the same. The tanks or vessels containing the active parts (core and windings) are indicated by dashed lines only.

In the figure P shows the loading of primary windings, the product of current and voltage for each of the coils. The primary windings for the lower stages are heavily loaded.

Each of the windings ‘2’ will carry a current of $I = P/V$

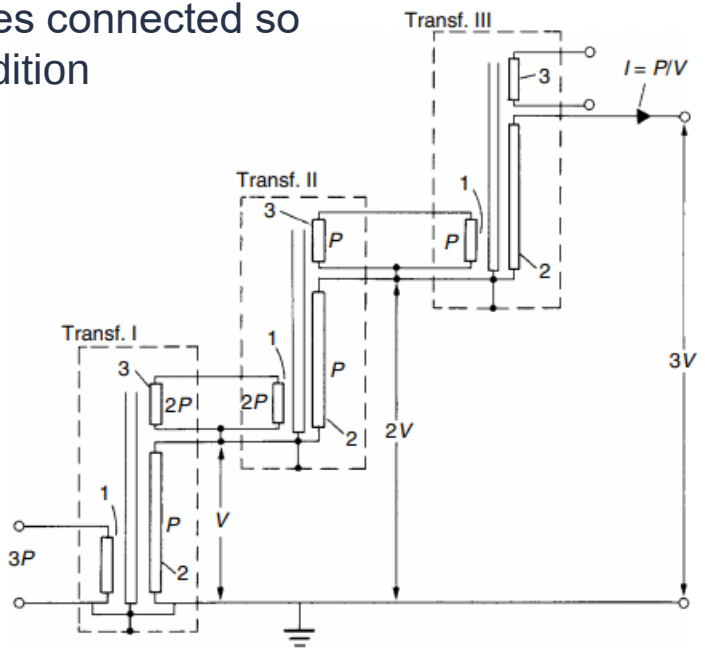


Figure 2.15 *Basic circuit of cascaded transformers. (1) Primary windings. (2) Secondary h.t. windings. (3) Tertiary exciting windings*

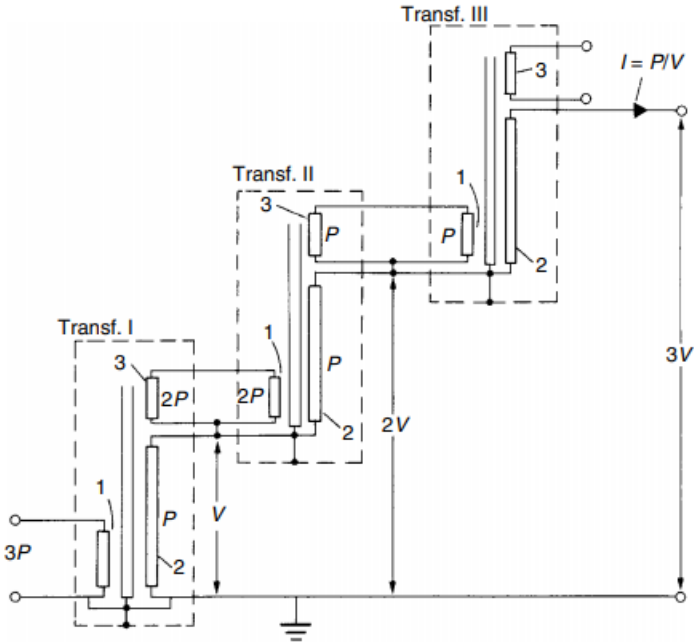
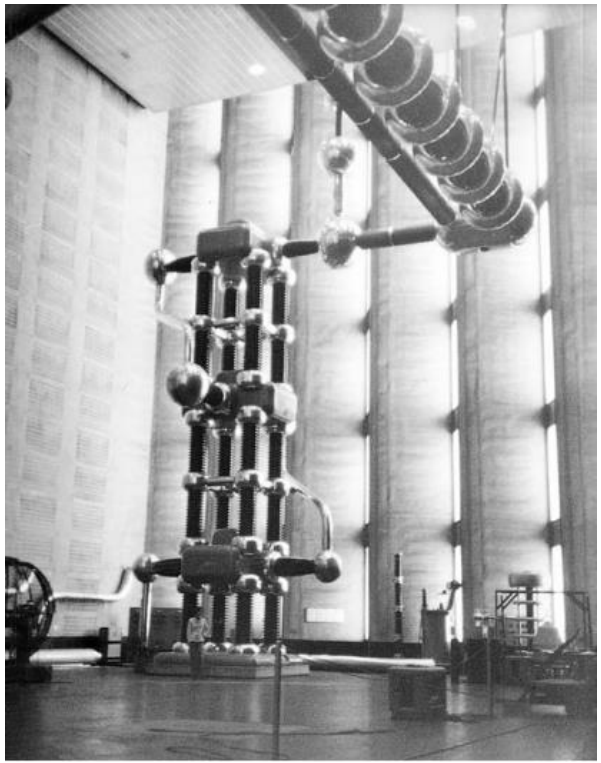


Figure 2.15 *Basic circuit of cascaded transformers. (1) Primary windings. (2) Secondary h.t. windings. (3) Tertiary exciting windings*



3 ALTERNATING VOLTAGES

The leakage or stray reactances X for each winding are the primary X_p, the secondary HV winding X_h and the exciting winding X_e. Neglecting losses within the windings and magnetizing currents, the simplified calculation of the resultant reactance of a cascade unit with n transformers:

$$X_{res} = \sum_{v=1}^{n} \left[X_h + (n-v)^2 X_e + (n+1-v)^2 X_p \right]$$

Assuming three equal transformer units, the equation leads to a resultant reactance of

$$X_{res} = 3X_h + 5X_e + 14X_p$$

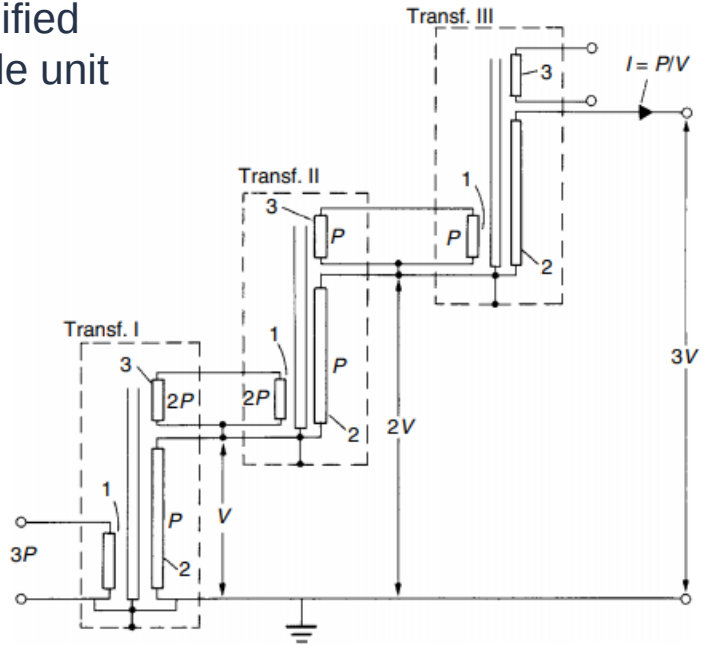


Figure 2.15 *Basic circuit of cascaded transformers. (1) Primary windings. (2) Secondary h.t. windings. (3) Tertiary exciting windings*

2.2 Series resonant circuits

The tuned series resonant HV testing circuit is developed for overcoming the accidental and unwanted resonance to which more conventional test sets are more prone.

If we consider a conventional straight test set system consisting of a transformer, testing a capacitor C, then its equivalent circuit will be $(r_1 + j\omega L_1)$ representing the impedance of the voltage regulator and the transformer primary, ωL represents the transformer shunt impedance, which is usually large compared with L_1 and L_2 and can be normally neglected. $(r_2 + j\omega L_2)$ represents the impedance of the transformer secondary. $1/\omega C$ represents the impedance of the load.

If by chance $\omega(L_1 + L_2) = 1/\omega C$, accidental resonance occurs. At supply frequency, the effect can be extremely dangerous as the instantaneous voltage application can be of the order of 20 times the intended high voltage.

This is highly possible to occur when testing at the maximum limit of current and relatively low voltage, i.e. High capacitive load.

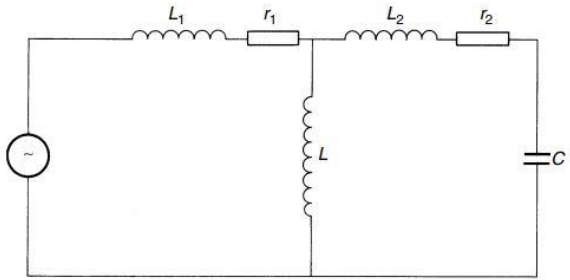


Figure 2.17 *Equivalent circuit of straight test set consisting of a transformer and test capacitance*

2.2 Series resonant circuits

With the series resonant set, however, the resonance is controlled at fundamental frequency and no unwanted resonance can therefore occur.

In each circuit shown in the figure, the capacitance C_t represents the almost pure capacitive load of the test objects and a constant power supply frequency is assumed.

In subfigure (a), a continuously variable inductance (reactor) is connected to the LV winding of a step-up trafo, whose secondary winding is rated for the full test voltage. In this way, the impedance of the reactor is converted to the HV side. If the inductance of the reactor is tuned to match the impedance of the capacitive load, the reactive power of the load is completely compensated.The step-up transformer, however, has to carry the full load current which is a disadvantage of this circuit.

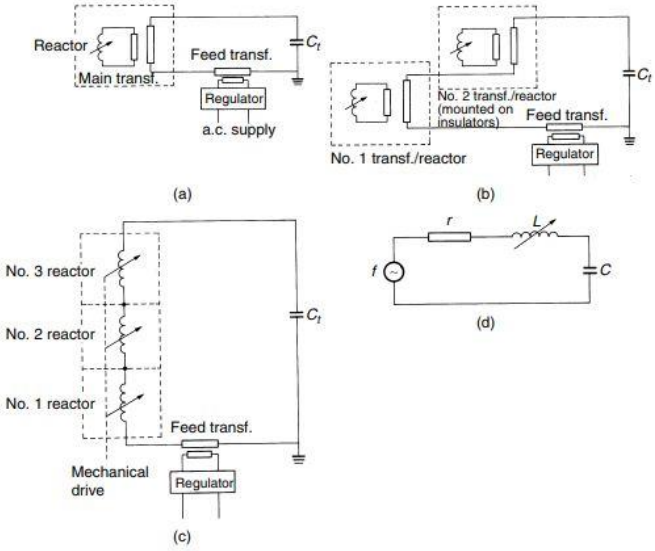


Figure 2.18 *Series resonant circuit for transformer/reactor. (a) Single transformer/reactor. (b) Two or more units in series. (c) Simplified diagram of s.r. circuit for h.t. reactor units in series. (d) Equivalent diagram of s.r. circuits*

The same disadvantage applies to circuit in subfigure (b), although no special means are necessary to cascade two or more units. The inductors are designed for high-quality factors $Q = \omega L/R$ within the limits of the inductance variation.

In subfigure (c), the inductance of the reactors may be changed by up to 10 to 20 times, offering the opportunity to tune the circuits with capacitances which vary within the same order.

The equivalent circuits for all these options is simply a low damped circuit shown in subfigure (d).

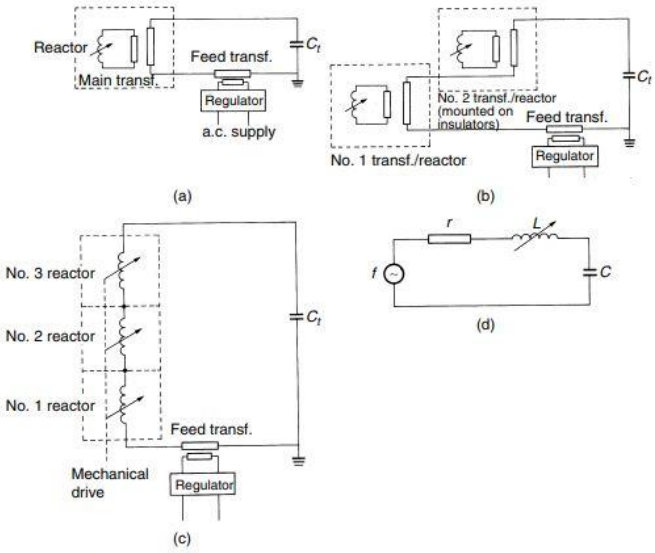


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3 ALTERNATING VOLTAGES

Additional advantages of the series resonant circuits:

- Voltage waveshape is improved not only by the elimination of unwanted resonances, but also by attenuation of harmonics already in the power supply.

- The power required from the supply is lower than the kVA in the main test circuit. (only 5% of the main kVA with a unity power factor).

- If a failure of the test specimen occurs, no heavy power arc will develop as only the load capacitance will be discharged. Additionally the arc will be self-extinguishing due to voltage collapse.
- The series or parallel operation of the HV reactor or LV reactor and HV transformer units is simple and very efficient.
- Auto-tuning devices can keep the set in tune, if supply frequency or load capacitance varies, or concerning auto-voltage control.

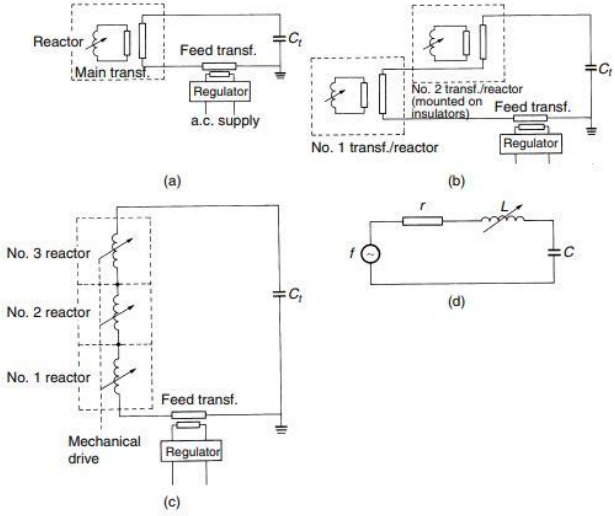


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As the load capacitance cannot be kept within narrow limits, the supply frequency can be continuously variable to achieve resonance. As shown in the figure, an exciter circuit connected to the LV mains,can be used to excite the circuit with a variable frequency, this supply is therefore designed as a controlled frequency converter, which are standard devices nowadays. This converter supplies again only the losses of the testing circuit, which are usually less than about 2% of the reactive power of C_t if frequencies equal to or higher than 50 Hz are used.

Due to resonance condition the frequency is always

$$f = \frac{1}{2\pi\sqrt{L_n C_t}}$$

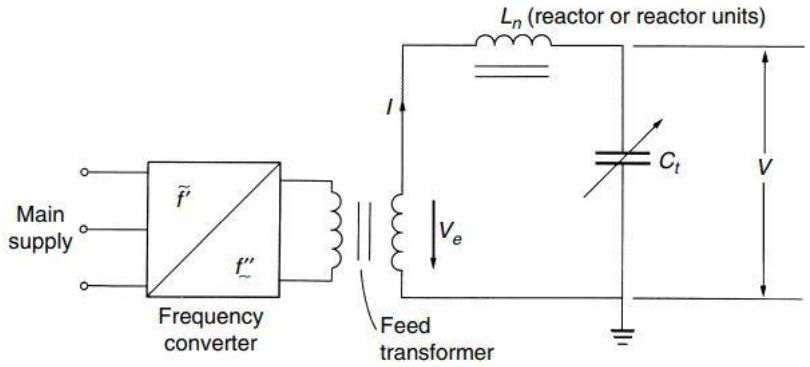


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

The nominal inductance L_n will be designed according to a nominal capacitance $C_n = C_t$ which is the highest capacitance that can be tested with the full rated voltage $V = V_n$ of the circuit and a nominal frequency f_n, which is the lowest frequency within this rated voltage.

$$L_n = \frac{1}{(2\pi)^2 f_n^2 C_n}$$

A further criterion for the design is the maximum or nominal current $I = I_n$, which either overheats the coil or saturates the iron core, if any. As the losses are very small, we may neglect $R \ll \omega L_n$.

I_n may directly be derived from the voltage drop across L_n

$$I_n = \frac{V_n}{2\pi f_n L_n} = V_n \sqrt{\frac{C_n}{L_n}}$$

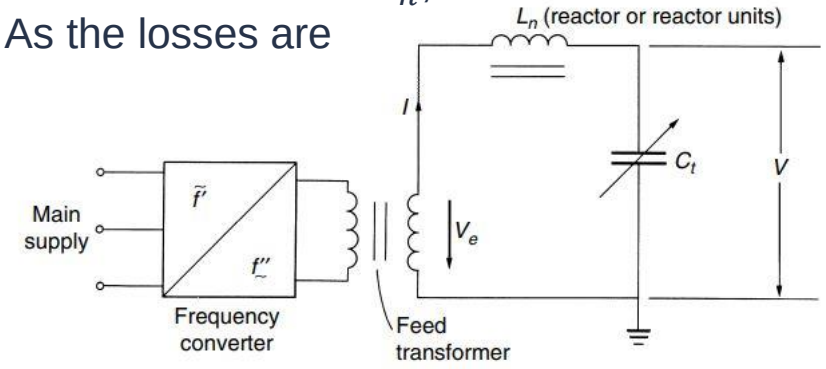


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

For test objects with capacitance values C_t different from C_n, the resulting testing frequency f will also be different from f_n. It can be calculated as

$$\frac{f}{f_n} = \sqrt{\frac{C_n}{C_t}} = \frac{1}{\sqrt{C_t/C_n}}$$

The below relation is present

$$\frac{I}{I_n} = \frac{f_n}{f} = \sqrt{\frac{C_t}{C_n}}$$

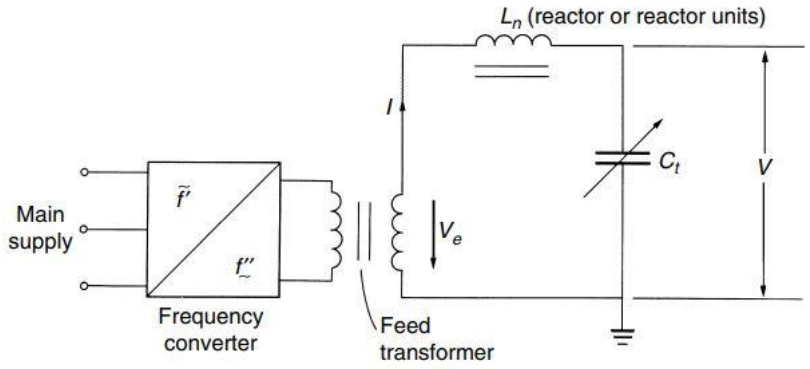


Figure 2.20 *Schematic diagram of s.r. test circuit with variable test frequency*

The normalized operating conditions are illustrated in the figure. For quite small test specimens, the test frequency f may conveniently be limited by the addition of a permanent capacitor as e.g. a capactive voltage divider; the relatively modest variation of this frequency for large capacitors under test will improve the flexibility of applications, i.e. for the testing of power cables with AC voltages.

The actual limitations in testing of very large test specimens with lower voltages than V_n are given by the reduction of Q for too low frequencies, and the frequency for which the exciter transformer saturates.

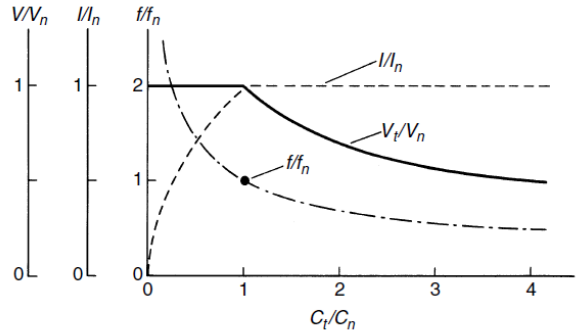


Figure 2.21 *Operating characteristics of circuit according to Fig. 2.20*

4 IMPULSE VOLTAGES

Disturbances of electric power transmission and distribution systems are frequently caused by two kinds of transient voltages, whose amplitudes may greatly exceed the peak values of the normal AC operating voltage:

- ligthning overvoltages.
- switching phenomena.

Lightning overvoltages originated by lightning strokes hitting the phase wires of overhead lines or the busbars of outdoor substations. The amplitudes are very high, usually in the order of 1000 kV or more, as very stroke may inject lightning current up to about 100 kA and even more into the tranmission line.

Each stroke is then followed by travelling waves, whose amplitude is often limited by the maximum insulation strength of the overhead line. The voltage rise of such a travelling wave is proportional to the steepnes of the lightning current, which may exceed 100 kA/μsec.

4 IMPULSE VOLTAGES

Too high voltage levels are immediately chopped by the breakdown of the insulation and therefore travelling waves with steep wave front and even steeper wave tails may stress the insulation of power transformers or other HV equipment severely. Lightning protection systems, surge arresters will damp and distort the travelling waves.

The switching amplitudes are always related to the operating voltage and the shape is influenced by the impedances of the system as well as by the switching conditions. The rate of voltage rise is usually slower but it is well known that the waveshape can also be very dangerous to insulation systems especially to atmospheric air insulation in tranmission systems with voltage levels higher than 245 kV.

Both types of overvoltages are also effective in the LV distribution systems as they are either produce by the usual switches or they have been transmitted from the HV systems. Here, they may often cause a breakdown of electronic equipment, as they can reach amplitudes of several kVs and testing of certain LV apparatus with transient voltages or currents is a need.

4 IMPULSE VOLTAGES

Although the actual shape of both kinds of overvoltages varies strongly, it became necessary to simulate these transient voltages by relatively simple means for testing purposes.

The various national and international standards define the impulse voltages as a unidirectional voltage which rises more or less rapidly to a peak value and then decays relatively slowly to zero.

IES Standard 60, widely accepted today, makes a distinction between lightning and switching impulses. Impulse voltages with front duration varying from less than one up to a few tens of microseconds are considered as lightning impulses.

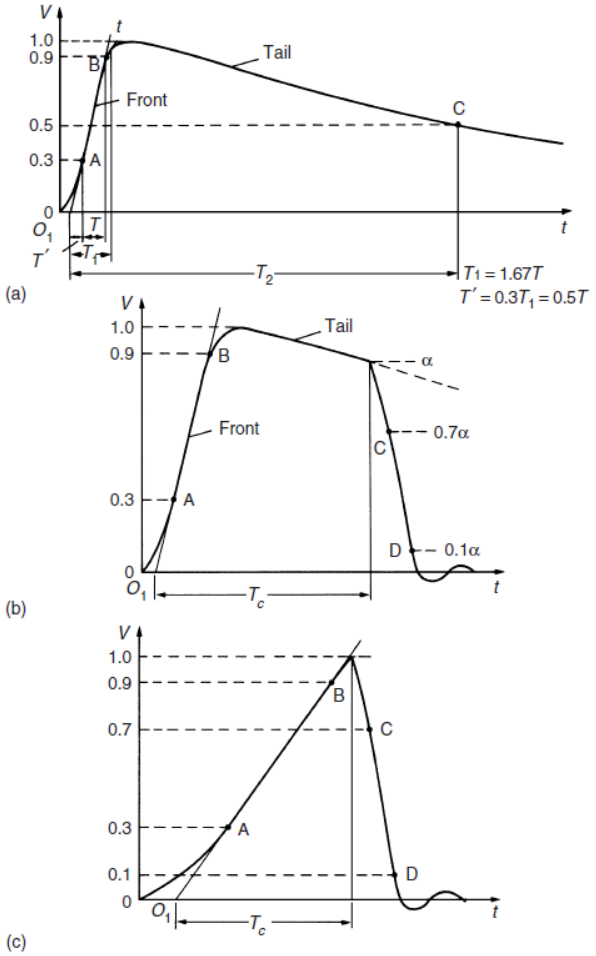


Figure 2.23 *General shape and definitions of lightning impulse (LI) voltages. (a) Full LI. (b) LI chopped on the tail. (c) LI chopped on the front. T_1: front time. T_2: time to half-value. T_c: time to chopping. O_1: virtual origin*

4 IMPULSE VOLTAGES

The ‘virtual origin’ O_1 is where the line AB cuts the time axis.

The ‘front time’ T_1 is defined as 1.67 times the interval between the instants when the impulse is 30% and 90% of the peak value.

The front time T_1 is usually 1.2 μs with $\pm 30\%$ tolerance and the time to half-value T_2 is 50 μs with $\pm 20\%$ tolerance.

1.2/50 impulse is the accepted standard lightning impulse voltage today.

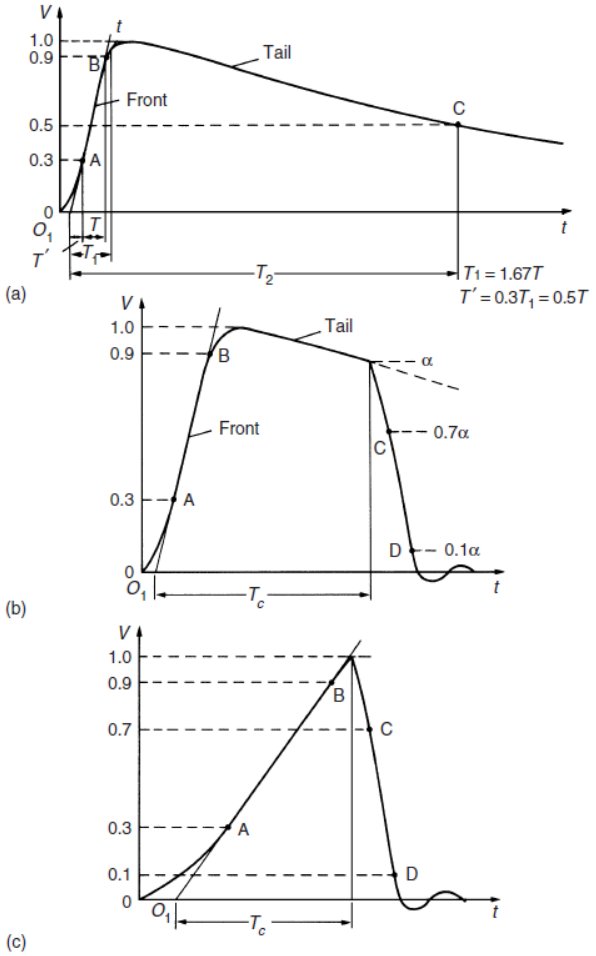


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4 IMPULSE VOLTAGES

The figure below shows the slope of a switching impulse. The time to half value T_2 is defined similarly as before, the time to peak T_p is the time interval between the actual origin and the instant when the voltage has reached its maximum value. However, it is difficult to etablish the actual crest value with high accuracy. An additional parameter time T_d the time at 90% of creast value.

The standard switching impulse has time parameters (and tolerances of):

$T_p = 250\mu s \pm 20\%$

$T_2 = 2500\mu s \pm 60\%$

and is therefore described as a 250/2500 impulse.

For investigating the insulation strength, the time to peak is usually varied between 100 and 1000 μs.

The breakdown strength of the insulation systems may be sensitive upon the voltage waveshape.

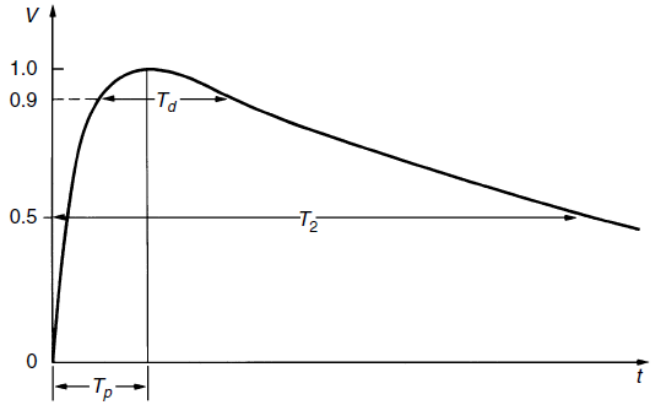


Figure 2.24 *General shape of switching impulse voltages. T_p: time to peak. T_2: time to half-value. T_d: time above 90 per cent*

IMPULSE VOLTAGES

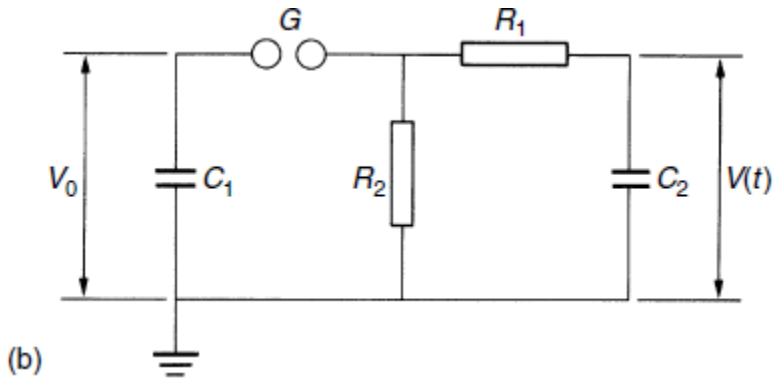
Single-stage impulse voltage generator circuits

In the circuit given in the figure, the capacitor C_1 is slowly charged from a DC source until the spark gap G breaks down. This spark gap acts as a voltage-sensitive switch, whose ignitiom time is very short in comparison to T_1. The resistors R_1 and R_2 and the capacitance C_2 form the waveshaping network.

R_1 will determine the front time T_1, R_2 will discharge the capacitors and control the wave tail. The test objects are connected in parallel to the C_2.

The maximum stored energy

$$W = \frac{1}{2} C_1 (V_{0max})^2$$



(b)

Multistage impulse voltage generator circuits

The difficulties encountered with spark gaps for the switching of very high voltages, the increase of the physical size of the circuit elements, difficulties suppressing corona discharges make the one-stage circuit inconvenient for higher voltages.

In order to overcome these difficulties, in 1923 Marx multistage generator has condensers charged in parallel through ohmic resistances and then dischargers in series through spark gaps. The DC voltage charges the equal stage capacitors C_1' in parallel through the resistors R' as well as the discharge resistance R_2'.

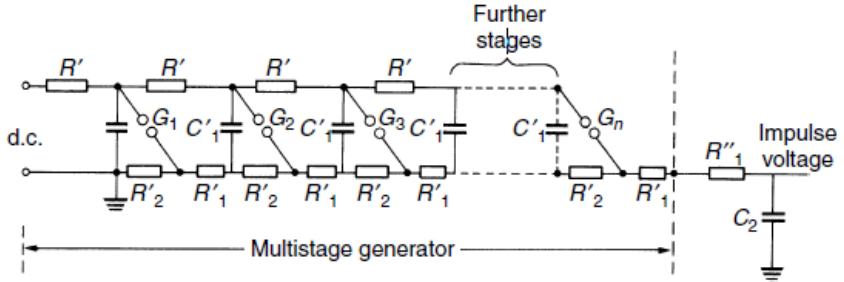


Figure 2.30 *Multistage impulse generator with distributed discharge and front resistors. R_2': discharge resistors. R_1': internal front resistors. R_1'': external front resistor*

IMPULSE VOLTAGES

Multistage impulse voltage generator circuits

At the end of the relatively long charging period (typically several seconds up to 1 minute), the discharge or firing of the generator is initiated by the breakdown of the lowest gap G_1 which is followed by a nearly simultanous breakdown of all the remaining gaps. In practice G_2 must be step to a gap distance a slightly higher than G_1.

Special triggering mechanism are used to ensure firing of the generator at the right time from the first stage.

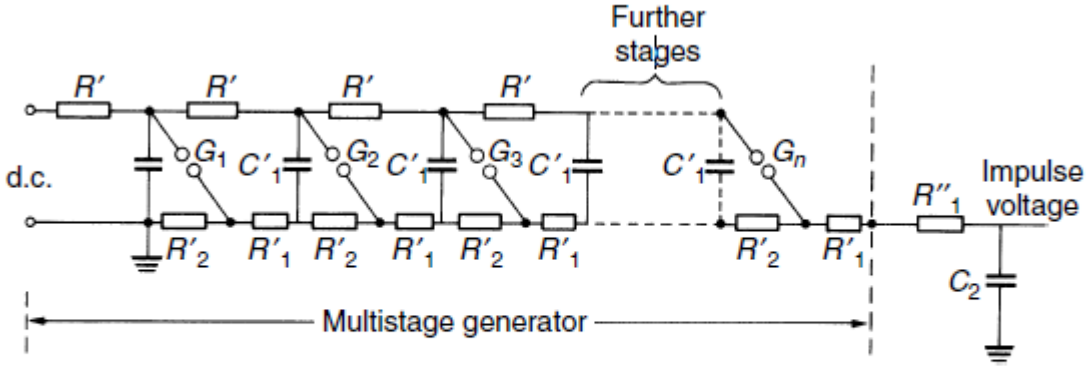


Figure 2.30 *Multistage impulse generator with distributed discharge and front resistors. R'_2: discharge resistors. R'_1: internal front resistors. R''_1: external front resistor*

4 IMPULSE VOLTAGES

Multistage impulse voltage generator circuits

A 20-stage, 4 MV, 200 kJ indoor impulse generator is shown in the figure.



END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm